EXERCISE 1

$Y_i \sim N(\beta_1 + \beta_2 e^{z_i}, 1)$ indep $i = 1, \dots, 120$

$Y_i \sim N(\beta_1 + \beta_3 e^{z_i^2}, 1)$ indep $i = 121, \dots, 200$

z_i known constants

a) yes:
1. normality, homoscedasticity, independence
2. the model is linear in $\beta_1, \beta_2, \beta_3$
3. the covariates are linearly independent.

b) sample space : $\mathcal{Y} = \mathbb{R}^{200}$

parameter space : $\Theta = \mathbb{R}^3$ (space of $(\beta_1, \beta_2, \beta_3)$)
σ^2 is known

c) the model can be written as

$Y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i \qquad i = 1, \dots, 200$

where we define

$\varepsilon_i \sim N(0, 1)$ iid

x_{i2} covariate $\quad x_{i2} = \begin{cases} e^{z_i} & \text{for } i = 1, \dots, 120 \\ 0 & \text{otherwise} \end{cases}$

x_{i3} covariate $\quad x_{i3} = \begin{cases} e^{z_i^2} & i = 121, \dots, 200 \\ 0 & \text{otherwise} \end{cases}$

in matrix form we get

$\underline{Y} = [Y_1, \dots, Y_{120}, Y_{121}, \dots, Y_{200}]^T$ vector of random variables (dim: 200×1)

$\underline{Y} \sim N_{200}(X\underline{\beta}, I_{200})$

X $(n \times p) = (200 \times 3)$ matrix of known constants

$$X = [\underline{1} \;\; \underline{x}_2 \;\; \underline{x}_3] = \begin{bmatrix} 1 & e^{z_1} & 0 \\ 1 & e^{z_2} & 0 \\ \vdots & \vdots & \vdots \\ 1 & e^{z_{120}} & 0 \\ 1 & 0 & e^{z_{121}^2} \\ \vdots & \vdots & \vdots \\ 1 & 0 & e^{z_{200}^2} \end{bmatrix}$$

$$\underline{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}$$ vector of unknown constants

$\underline{\varepsilon} = [\varepsilon_1, \dots, \varepsilon_{120}, \varepsilon_{121}, \dots, \varepsilon_{200}]^T$ vector of random variables

$\underline{\varepsilon} \sim N_{200}(\underline{0}, I)$ $\qquad I$ 200×200 identity matrix

d)

$$X^T X = \begin{bmatrix} \underline{1}^T \\ \underline{x}_2^T \\ \underline{x}_3^T \end{bmatrix} \cdot [\underline{1} \;\; \underline{x}_2 \;\; \underline{x}_3] = \begin{bmatrix} \underline{1}^T\underline{1} & \underline{1}^T\underline{x}_2 & \underline{1}^T\underline{x}_3 \\ \underline{x}_2^T\underline{1} & \underline{x}_2^T\underline{x}_2 & \underline{x}_2^T\underline{x}_3 \\ \underline{x}_3^T\underline{1} & \underline{x}_3^T\underline{x}_2 & \underline{x}_3^T\underline{x}_3 \end{bmatrix}$$

$$= \begin{bmatrix} 200 & \sum_{i=1}^{120} e^{z_i} & \sum_{i=121}^{200} e^{z_i^2} \\ \sum_{i=1}^{120} e^{z_i} & \sum_{i=1}^{120} e^{2z_i} & 0 \\ \sum_{i=121}^{200} e^{z_i^2} & 0 & \sum_{i=121}^{200} e^{2z_i^2} \end{bmatrix}$$

$$\underset{3\times200 \;\; 200\times1}{X^T \underline{y}} = \begin{bmatrix} \underline{1}^T \\ \underline{x}_2^T \\ \underline{x}_3^T \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_{200} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{200} y_i \\ \sum_{i=1}^{120} e^{z_i} y_i \\ \sum_{i=121}^{200} e^{z_i^2} y_i \end{bmatrix}$$

the MLE $\hat{\underline{\beta}}$ is found as $\hat{\underline{\beta}} = (X^T X)^{-1} X^T \underline{y}$

e) the distribution of $\hat{\underline{\beta}}(\underline{Y})$ is $\hat{\underline{\beta}}(\underline{Y}) \sim N_3(\underline{\beta}, \underbrace{(X^T X)^{-1}}_{(X^T X)^{-1} \underset{1}{\sigma^2}})$

f) $\underline{e} = \underline{y} - X\hat{\underline{\beta}}$

$\sum_{i=1}^{200} e_i = 0$ yes: the model includes the intercept

$\sum_{i=1}^{200} e_i z_i = 0$ no: $[z_1, \dots, z_{200}]^T$ does not belong to $C(X)$ (column space of X)

$\sum_{i=1}^{200} e_i z_i^2 = 0$ no

$\sum_{i=1}^{200} e_i e^{z_i} = 0$ no

$\sum_{i=1}^{200} e_i e^{z_i^2} = 0$ no

$\sum_{i=1}^{120} e_i e^{z_i} = 0$ yes

EXERCISE 2

$n = 100$

$$CHD_i = \begin{cases} 1 & \text{if individual } i \text{ has heart disease} \\ 0 & \text{otherwise} \end{cases}$$

(a1) The response variable $CHD_i \in \{0,1\}$ is binary, so I can fit a logistic regression (GLM for Bernoulli data based on the logit link)

- $CHD_i \sim \text{Bernoulli}(\pi_i)$ independent for $i = 1, \dots, 100$
- linear predictor $\eta_i = \underset{\sim}{x}_i^T \underline{\beta} = \beta_1 + \beta_2 AGE_i$
- $\log\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i$ logit link function

$$\Rightarrow \text{logit}(\pi_i) = \log \frac{\pi_i}{1-\pi_i} = \beta_1 + \beta_2 AGE_i$$

(a2) interpretation of AGE coefficient β_2

Consider two individuals i with covariate AGE_i

j with covariate $AGE_j = AGE_i + 1$

$$\text{logit}(\pi_i) = \beta_1 + \beta_2 AGE_i$$

$$\text{logit}(\pi_j) = \beta_1 + \beta_2 AGE_j = \beta_1 + \beta_2 (AGE_i + 1) = \beta_1 + \beta_2 AGE_i + \beta_2$$

$$\beta_2 = \text{logit}(\pi_j) - \text{logit}(\pi_i) = \log \underbrace{\frac{\pi_j}{1-\pi_j}}_{\text{odds for individual } j} - \log \underbrace{\frac{\pi_i}{1-\pi_i}}_{\text{odds for individual } i}$$

If I increase the age of 1 year, the log-odds of having a heart disease increase of 0.1109.

Equivalently

$$\beta_2 = \log \frac{\frac{\pi_j}{1-\pi_j}}{\frac{\pi_i}{1-\pi_i}} \Rightarrow e^{\beta_2} = \frac{\frac{\pi_j}{1-\pi_j}}{\frac{\pi_i}{1-\pi_i}}$$

$$\Rightarrow \frac{\pi_i}{1-\pi_i} \cdot e^{\beta_2} = \frac{\pi_j}{1-\pi_j}$$

The odds change of a multiplicative factor $e^{0.1109}$ if I increase the age of 1 year

(a3) $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$

the test is based on the statistic $z = \frac{\hat{\beta}_2}{\hat{se}(\hat{\beta}_2)} = 4.61$ (in the table)

since the p-value is ~ 0, I reject H_0.

Hence, $\beta_2 \neq 0$, which means that age affects the probability of having a heart disease.

(b1)

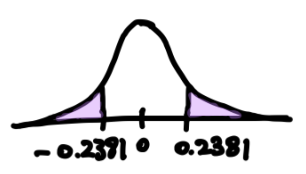
- $CHD_i \sim \text{Bernoulli}(\pi_i)$ independent for $i = 1, \dots, 100$
- linear predictor $\eta_i = \underset{\sim}{x}_i^T \underline{\beta} = \beta_1 + \beta_2 AGE_i + \beta_3 AGE_i^2$
- $\log\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i$ logit link function

$$\Rightarrow \text{logit}(\pi_i) = \log \frac{\pi_i}{1-\pi_i} = \beta_1 + \beta_2 AGE_i + \beta_3 AGE_i^2$$

(b2)

- estimate of the intercept: $z_1^{obs} = \frac{\hat{\beta}_1}{\hat{se}(\hat{\beta}_1)} \Rightarrow \hat{\beta}_1 = z_1^{obs} \cdot \hat{se}(\hat{\beta}_1) = -0.99 \cdot 4.2901 = -4.2472$
- estimate of the age coeff.: $z_2^{obs} = \frac{\hat{\beta}_2}{\hat{se}(\hat{\beta}_2)} \Rightarrow \hat{\beta}_2 = z_2^{obs} \cdot \hat{se}(\hat{\beta}_2) = 0.315 \cdot 0.1947 = 0.0613$
- z-value of age^2: $z_3^{obs} = \frac{\hat{\beta}_3}{\hat{se}(\hat{\beta}_3)} = 0.2381$

$$\text{pvalue} = \mathbb{P}_{H_0}(|z_3| \geq |z_3^{obs}|) = 2 \cdot \mathbb{P}_{H_0}(z_3 \geq |z_3^{obs}|) = 2\underbrace{(1 - \Phi(0.2391))}_{>0.10} > 0.20$$



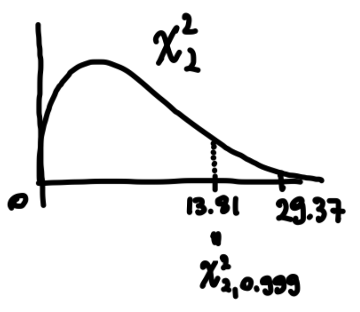
(b3) $\begin{cases} H_0: \beta_2 = \beta_3 = 0 \\ H_1: \text{at least one of } \beta_2, \beta_3 \text{ is} \neq 0 \end{cases}$

likelihood ratio test

$$W = 2 \log \frac{\hat{L}(\text{model})}{\tilde{L}(\text{null model})} = 2\{\hat{\ell}(\text{model}) - \tilde{\ell}(\text{null model})\} \; \dot\sim \; \chi^2_2 \text{ under } H_0$$

$$w^{obs} = D(\text{null}) - D(\text{model}) = 136.66 - 107.29 = 29.37$$

$\mathbb{P}_{H_0}(W \geq w^{obs}) < 0.001$ I reject H_0



(b4) $\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases}$

We already performed the test in (b2). We do not reject H_0.

Hence we prefer model (a).

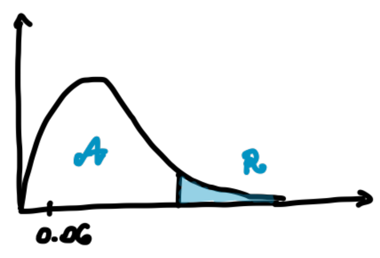
We can equivalently use the deviance to compare nested models

$$W = 2\{\hat{\ell}(\text{model } b) - \hat{\ell}(\text{model } a)\} \; \dot\sim \; \chi^2_1$$

$$w^{obs} = 2\{\tilde{\ell}(\text{saturated}) - \hat{\ell}(\text{model } a) - \tilde{\ell}(\text{saturated}) + \hat{\ell}(\text{model } b)\}$$

$$= D(\text{model } a) - D(\text{model } b) = 107.35 - 107.29 = 0.06$$

I do not reject H_0



(c1) the new variable $z_i = AGE < 50$ is $= \begin{cases} 1 & \text{if } AGE_i < 50 \\ 0 & \text{if } AGE_i \geq 50 \end{cases}$

- $CHD_i \sim \text{Bernoulli}(\pi_i)$ independent for $i = 1, \dots, 100$
- linear predictor $\eta_i = \underset{\sim}{x}_i^T \underline{\beta} = \beta_1 + \beta_2 z_i$
- $\log\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i$ logit link function

$$\Rightarrow \text{logit}(\pi_i) = \log \frac{\pi_i}{1-\pi_i} = \beta_1 + \beta_2 z_i$$

(c2) I consider individual i with $AGE_i \geq 50$ ($\Rightarrow z_i = 0$)

individual j with $AGE_j < 50$ ($\Rightarrow z_i = 1$)

$$\left.\begin{aligned} \log \frac{\pi_i}{1-\pi_i} &= \beta_1 \\ \log \frac{\pi_j}{1-\pi_j} &= \beta_1 + \beta_2 \end{aligned}\right\} \Rightarrow \beta_2 = \log \frac{\pi_j}{1-\pi_j} - \log \frac{\pi_i}{1-\pi_i}$$

$$= \log \left\{ \frac{\frac{\pi_j}{1-\pi_j}}{\frac{\pi_i}{1-\pi_i}} \right\} = \log \left\{ \frac{\frac{\mathbb{P}(Y_i = 1 \mid AGE_i < 50)}{\mathbb{P}(Y_i = 0 \mid AGE_i < 50)}}{\frac{\mathbb{P}(Y_i = 1 \mid AGE_i \geq 50)}{\mathbb{P}(Y_i = 0 \mid AGE_i \geq 50)}} \right\} = -2.0989$$

The odds of having a coronary heart disease of an individual older than 50 y.o. are multiplied by -2.0989 to obtain the odds of an individual younger than 50 y.o.